

Context Management in Large Language Models: A Literature Review of Human-Inspired Memory Architectures

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Abstract

This literature review surveys the landscape of context management techniques in large language models (LLMs), with a particular focus on architectures inspired by human memory systems. We examine how short-term, medium-term, and long-term memory analogues have been developed to address the fundamental context window limitations of transformer-based models. Spanning attention mechanisms, cache optimization, memory augmentation, and evaluation methodologies, this review synthesizes over 200 works to provide a comprehensive reference for researchers working at the intersection of cognitive science and natural language processing.

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